

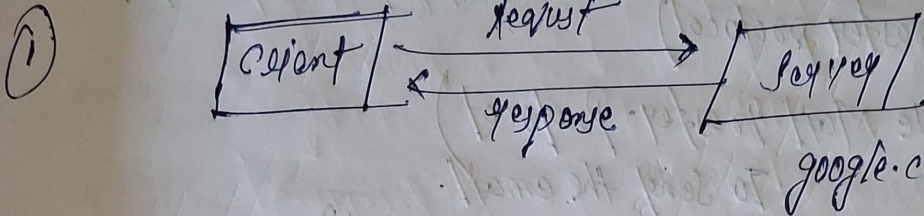
Application Layer.

Module - IV

T-1:30:20

The topmost layer of TCP/IP or OSI model that provides network services directly to user applications and enables communications b/w software applications over a network.

→ whatsapp, files, Email, web browser.



Data center - collection of servers in big companies are called data center.

→ collection of huge number of computers.

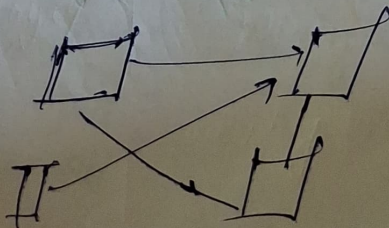
→ Static IP Addresses, that do not change.

Ping - It measures the round trip time for messages sent from originating host to destination of the computer. And that are echo back. Entire time is Ping time.

can we reduce ping time?

No,

② Peer to Peer: Various computers are connected with each other, there is no one dedicated server. Every single computer can form as a client or a server.



Protocols → web protocol.

TCP/IP:

HTTP → How data is transfer

DHCP (Dynamic Host control protocol)

↳ allocate IP add to your device that connected to network.

FTP (File Transfer protocol)

SMTP (Simple Mail Transfer Protocol)

To send the email.

POP3 & IMAP (This for received the Email)

SSH (Secure Shell) if I want to login someones computer /
terminal of their computer we use (SSH)

VNC (Virtual Network control)

Telnet & VDP.

How do that Application talk to each other



Program : whatsapp / Email / Facebook
(one of the softw. I can use)

↳ same work happen.

↓
Sending a message

↓
Record video.

↓
Refresh the page

↓
open camera.

Thread
(only one single job
can do at a time)

Socket

IP Address : It tell about which device we are working with

Port : which Application in that device we are working with

that can possible many tabs are open

in my device → chrome window open / whatsapp / spotify

So, I want data from my friend

how do I get which particular Tab will receive that data?

All using same IP Address.

Ephemeral port → Externally / Temporary (your PC)

it take a temporary port number automatically.

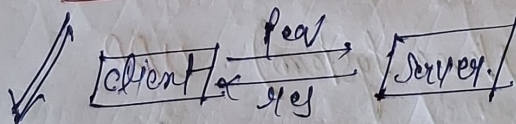
assigned by the operating system to the client application for the duration of a network connection.

• Released After the connection ends.

* HTTP protocol

→ Persistent : Same TCP connection reuse for every multiple object

→ Non-persistent : New TCP connect every time.



It is a client server protocol which tells us how you req data from server and it also tells us how the data server send the data to server.

Appli Layer

→ HTTP use TCP → Transport Layer

↳ Stateless (Every time you go, think new user)

HTTP method

Get : taking data

Post : giving data

put

del

Status code

2xx - Information

2xx - Success

3xx - redirection

4xx - client error

5xx - Server error

every is copy

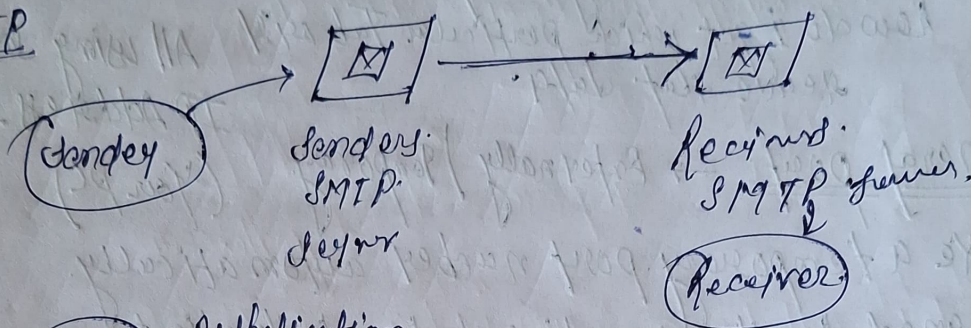
Cookies

(follow is copy)

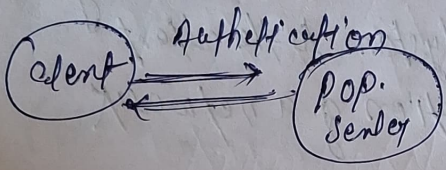
- unique storing
- stored in my browser.

How Email work (TCP) SMTP, POP3, IMAP.

SMTP



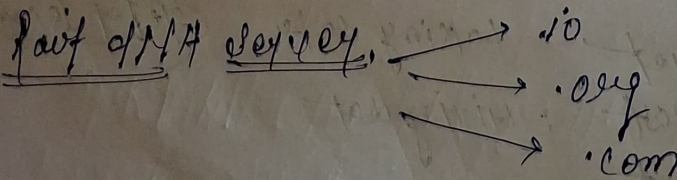
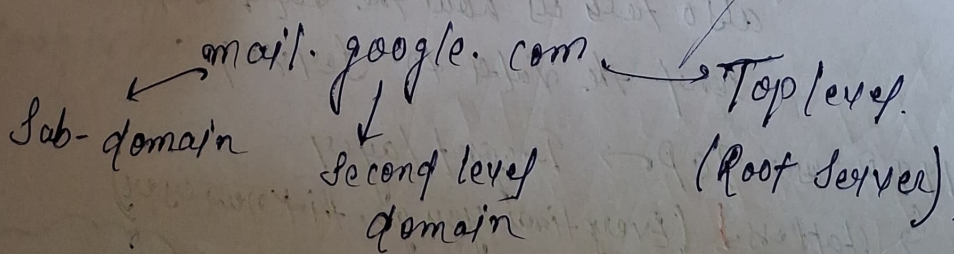
POP



IMAP: Internet Message Access protocol.
It allow to view your email multi devices.

* DNS (Domain Name System) UDP

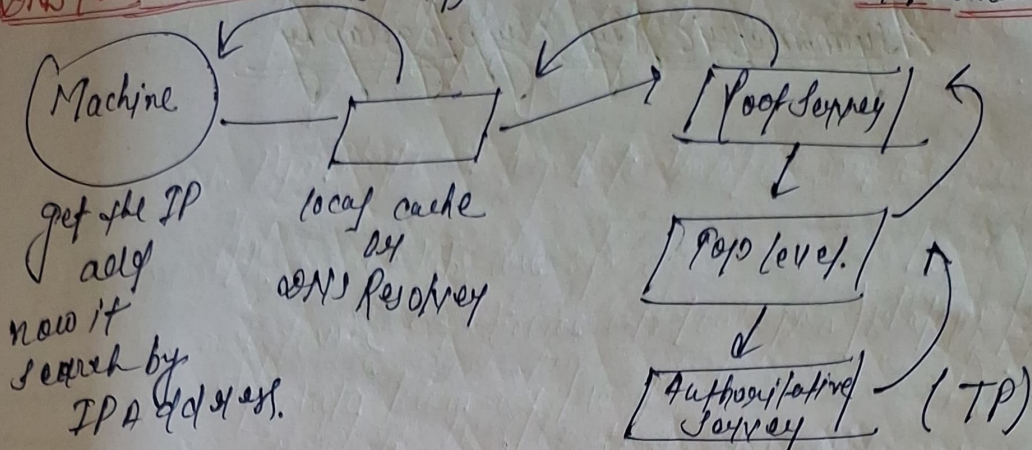
It is a naming system that translate a domain name to its IP address.



DNS (work flow).

(IP Addr)

Gate Smasher - L81



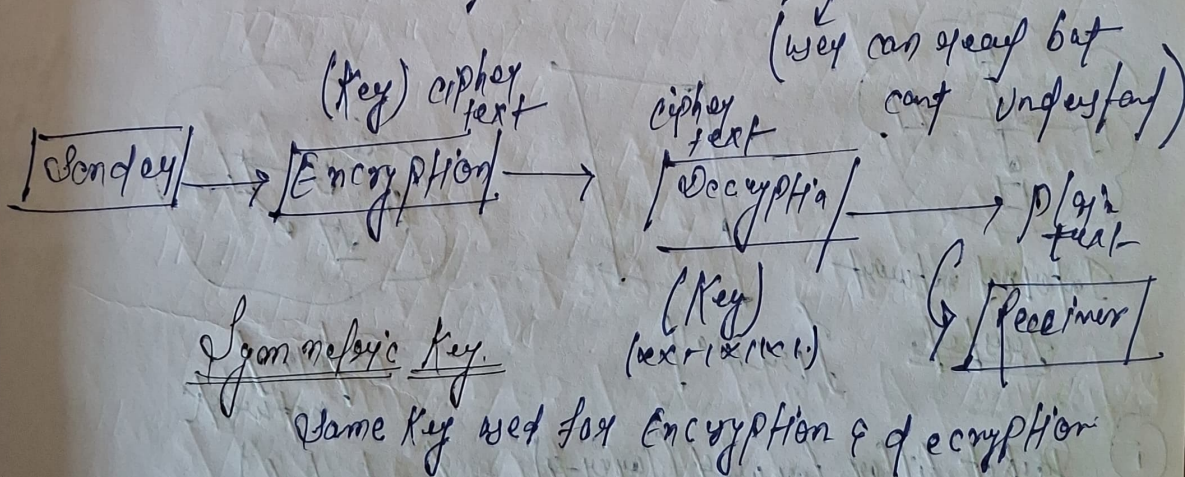
Node → Socket Programming L-86.

It is a way that how application will be communicating with each other over the internet.

Cryptography

- Protects sensitive data.
- Ensures secure communication.
- Prevents unauthorized access.

Cryptography is the process/technique through which we convert plain text to cipher text.



Sender → Message → Receiver

$$C = [K_1][M]$$

$$M = [K_2][C]$$

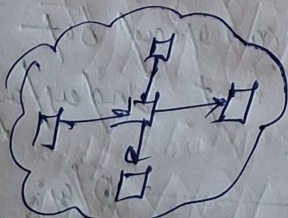
K_2 for cipher text Key

Asymmetric Key cryptography

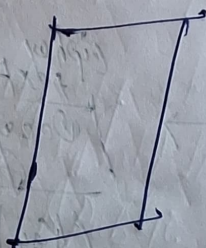
Firewalls (Network Security)

It is a device which Monitors and controls Incoming and outgoing traffic based on predefined rules.

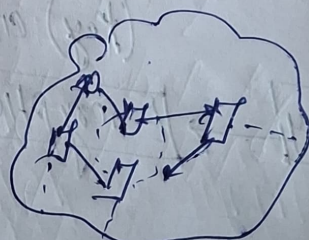
- Acts like a barrier
- Host based and Network based Firewall



Internal network
Firewall



Firewall



External network
(untrusted).

(i) Packet Filter Firewall (layer-4)

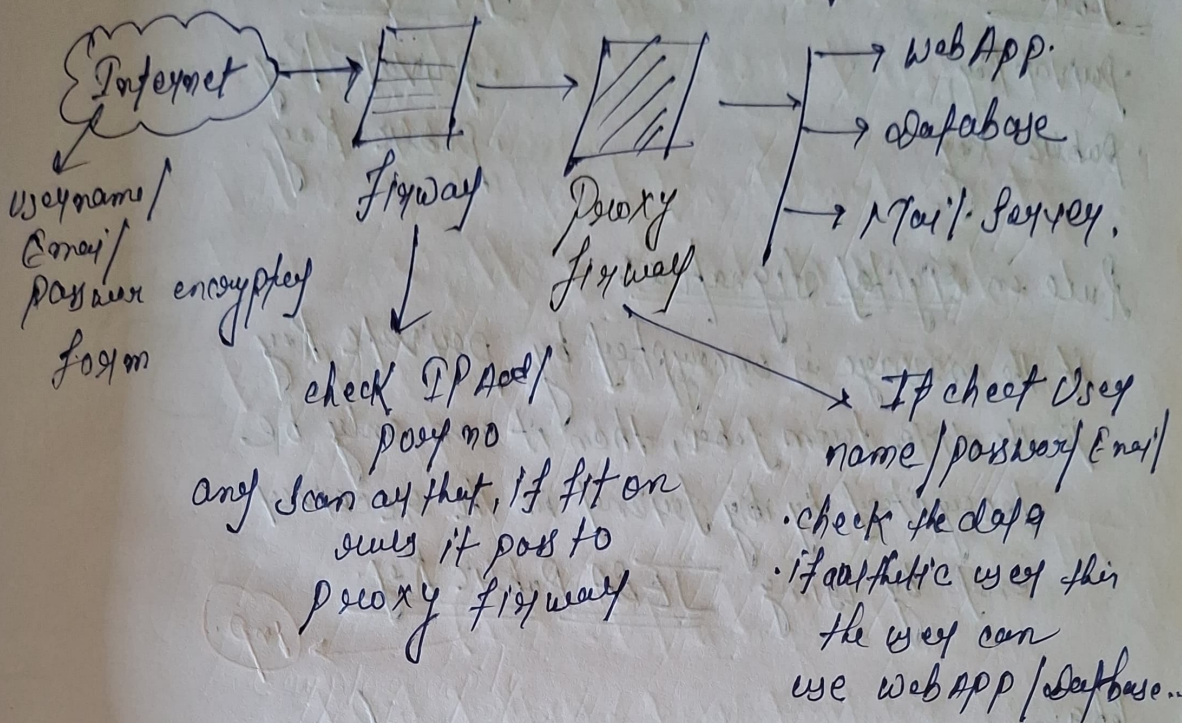
- check IP header, TCP header
- work on network and Transport layer
- can block IP add, full network → can block service (RDP, etc)

(ii)

Application (Proxy Firewall) Layer-5

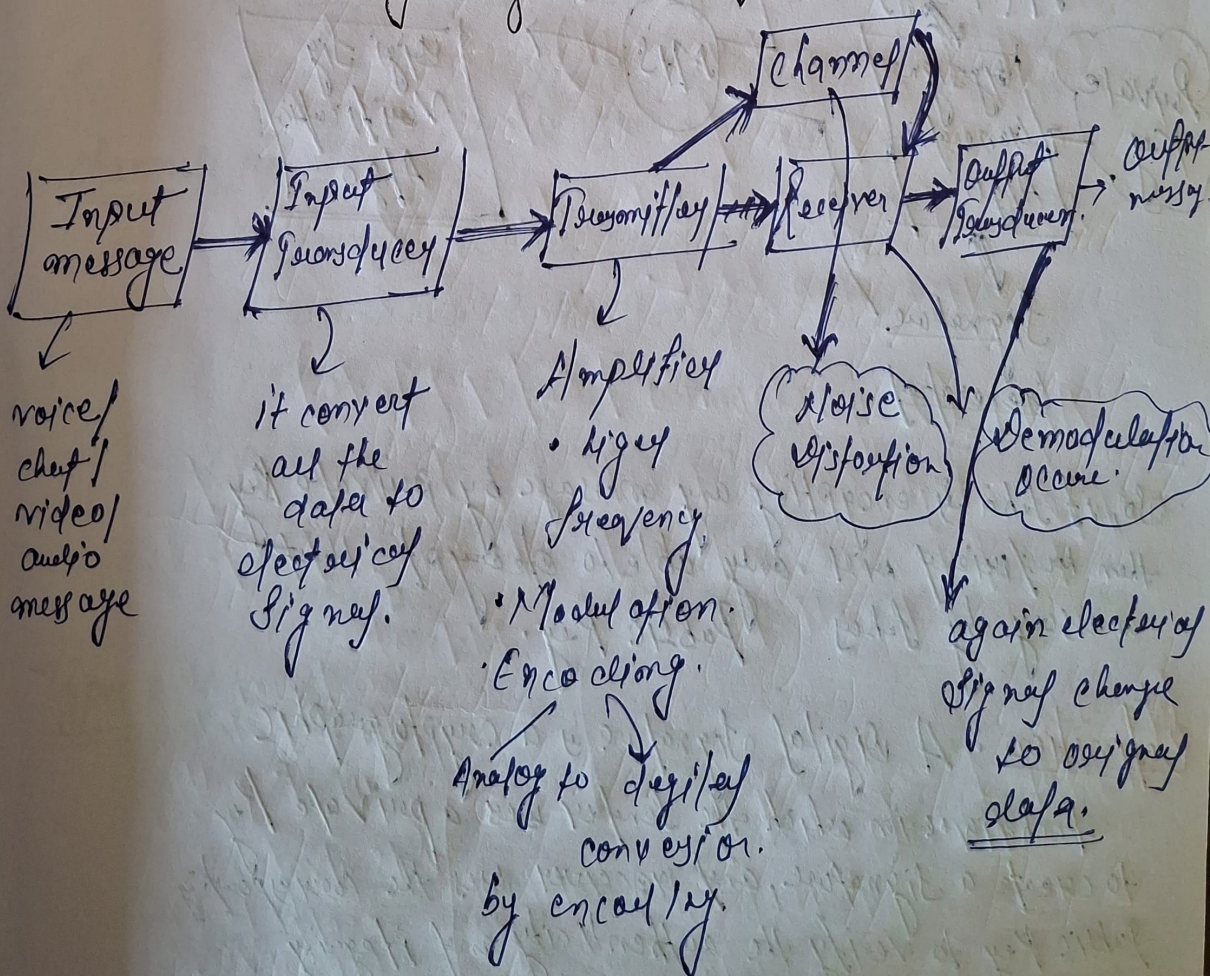
Lec-94

Proper Username/Email



Block Diagram of Communication.

L-97



Digital Signature.

by S-mint Engineer.

Seba.

- Private Key.
- Public Key

Sevend

- Private Key.
- Public Key.

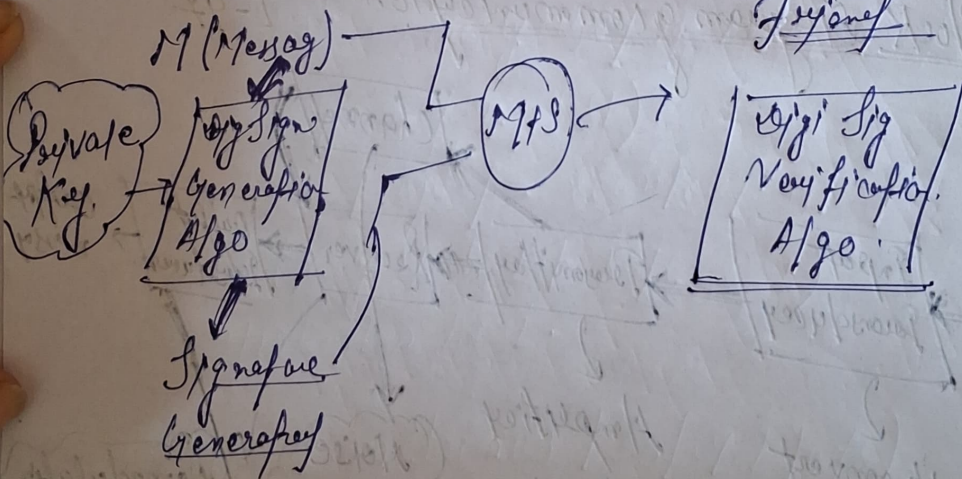
Rule in Digital Signature:

if the message is encrypted by private key in the side from seba, then it can only be decrypted by my sevend in the receiver end by using TEBA Public Key.

(imp.)

Seba.

Sevend



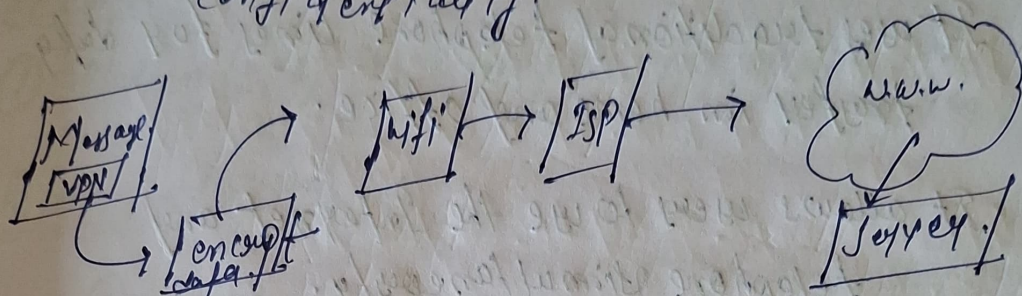
So, if seba is encrypting any message by private key then the sevend will only able to decrypt that message by seba's Public Key.

Definition: A Digital Signature is a cryptographic technique in which the sender uses a private key to create a signature, and receiver uses the corresponding public key to verify the authenticity and integrity of the message.

VPN (Virtual Private Network)

→ It is used to Establish a secure & reliable connection over an insecure network (Internet)

→ It will encrypt the data and provide confidentiality.



ISDN (Integrated Service Digital Network)

It is a digital communication technology that allows transmission of voice, data, text and more over a single telephone network.

Channels in ISDN

B channel (Bearer channel)

- carries voice, data and video.
- Data rate = 64 Kbps.

D channel (Delta channel)

- use for control Information.

Data Rate = 16 Kbps (BRI) or 64 Kbps (PRI)

Types of ISDN

BRI (Basic Rate Interface)

- used for homes and small offices
- structure BTD

PRI (Primary Rate Interface)

- used in large organization / Spent - 30BTS

DSL (Digital Subscriber Line)

✓ DSL is a technology that provides high-speed Internet access using existing telephone lines.

It uses traditional telephone lines for data transfer in home and office.

It allows users to use the Internet and telephone simultaneously.

Types of DSL:

HDSL (Asymmetric DSL)

- Download speed > upload speed.
- Most commonly used in home.

SDSL (Symmetric DSL)

- upload speed = download speed

VDSL (Very High Bit Rate DSL)

- Higher than ADSL.

ATM (Asynchronous Transfer Mode)

ATM is a high-speed networking technology that transfers voice, vid, and data using fixed-size packet called cell.

Features:

- High speed communication
- Uses fixed size (53 bytes)

Header (5 bytes) → data (48 bytes)

- Support voice, vid, data transfer.

Advantages

- Low transmission delay
- Efficient data transfer
- Support multiple services.

Disadvantages

- complex technology
- Expensive
- less commonly used today.

Wireless LAN (IEEE 802.11)

It is a standard for WLAN (Wireless Local Area Network) commonly known as Wi-Fi, allow device to communicate wirelessly using radio wave.

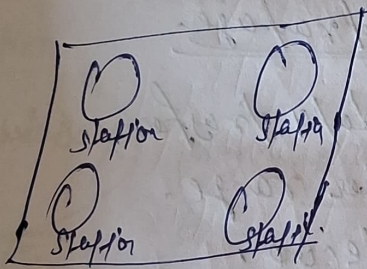
It consist of:

- ① Station (STA) ② Access Point (AP) ③ Distribution System (DS)

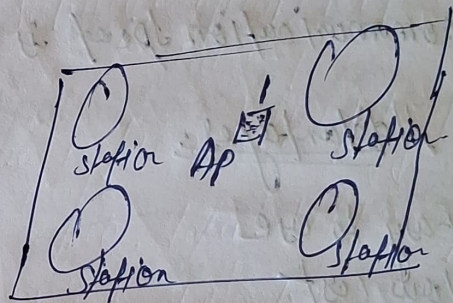
Architecture Diagram # Makaut

- ① BSS (Basic Service Set) ② ESS (Extended Service Set)

Basic Service Set:-

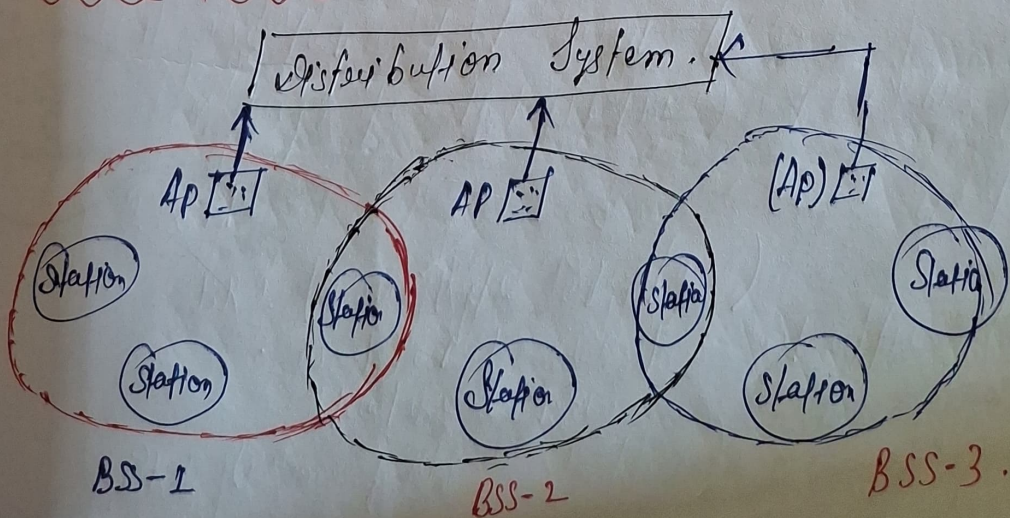


All the station communicate without Access point (AP) is **ADHOC NETWORK**.



Here we (AP) so if any station want to communicate they have to via Access point (AP), **INFRASTRUCTURE NETWORK**.

② ESS (Extended Service Set):-



Blue-Tooth.

By perfect computer Engineer

Bluetooth is a wireless local Area Network (WLAN) technology used for exchanging data over smaller distance.

- (i) Bluetooth has a limited coverage area of about 10m. No need any infrastructure.
- (ii) It is a example of ad-hoc network.
- (iii) Compared to wifi with bluetooth $\rightarrow$ Slow, limited range.
- (iv) communication speed is 12 mbps.

Advantages:

- Easy to use
- low cost
- wireless technology

Disadvantage:

- Slow data transfer.
- smaller range.
- less secure.

